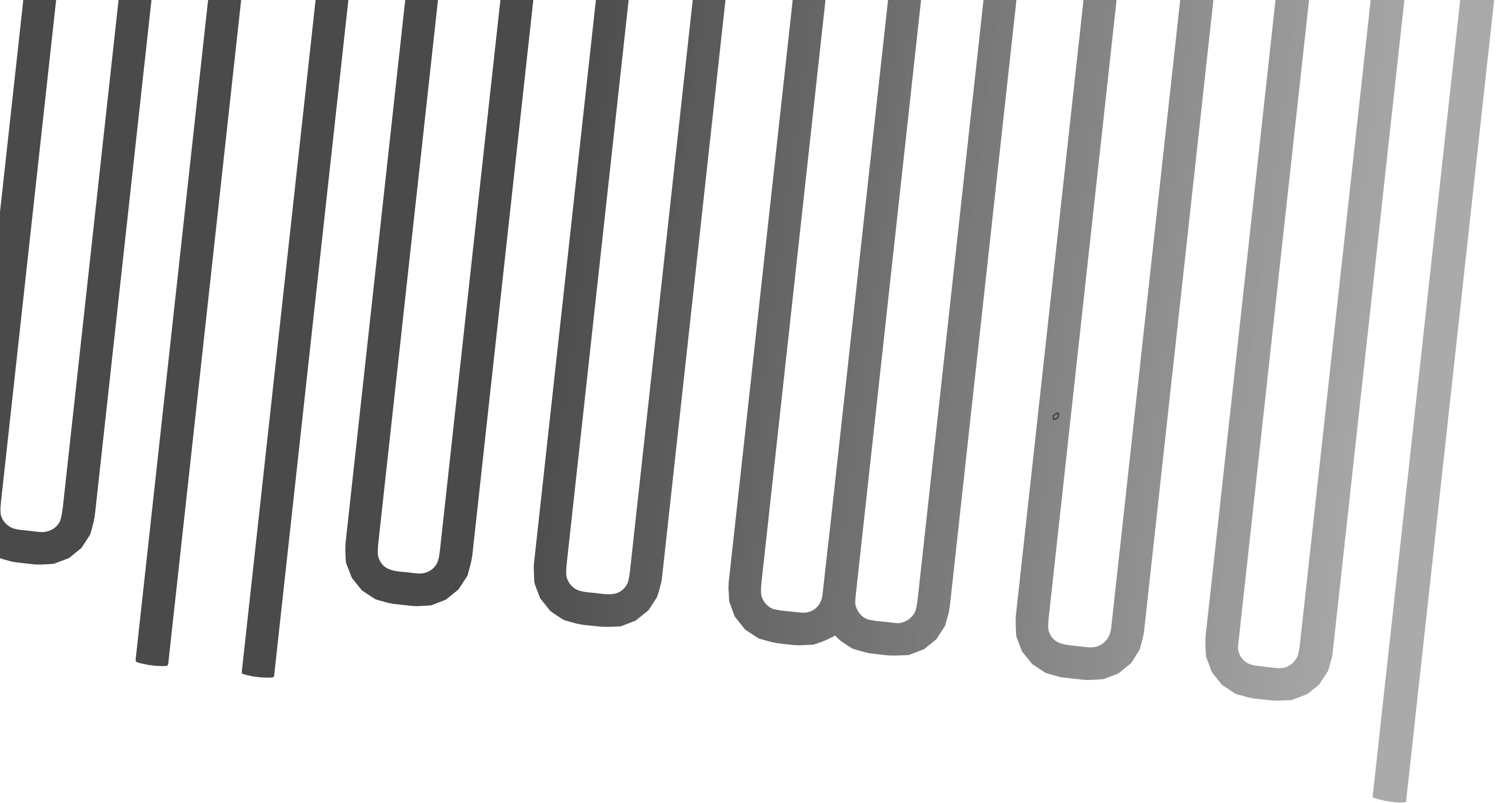




THERMOFLUIDS: BATTERY

Assignment 2-2

User Problems

The lonely off-grid community

Intended User

Battery design is driven by the user's needs and constraints, so it's important to first define who the system is for.

Green Gary



Lives off-grid, generating electricity from solar panels and a small wind turbine. Energy is stored in a home battery and used when generation is low or unavailable. This makes the battery the primary source of daily power, so reliability is critical, with performance and efficiency directly affecting how much usable energy is available.



Long Lifespan

The battery is used every day, so it must last for years under repeated charge and discharge. [12]

- Higher temperatures speed up degradation and reduce capacity over time.
- Uneven ageing makes the system less reliable, as weaker parts limit overall performance.



Energy Efficient

Energy is limited to what is generated and stored, especially during low wind or sunlight.

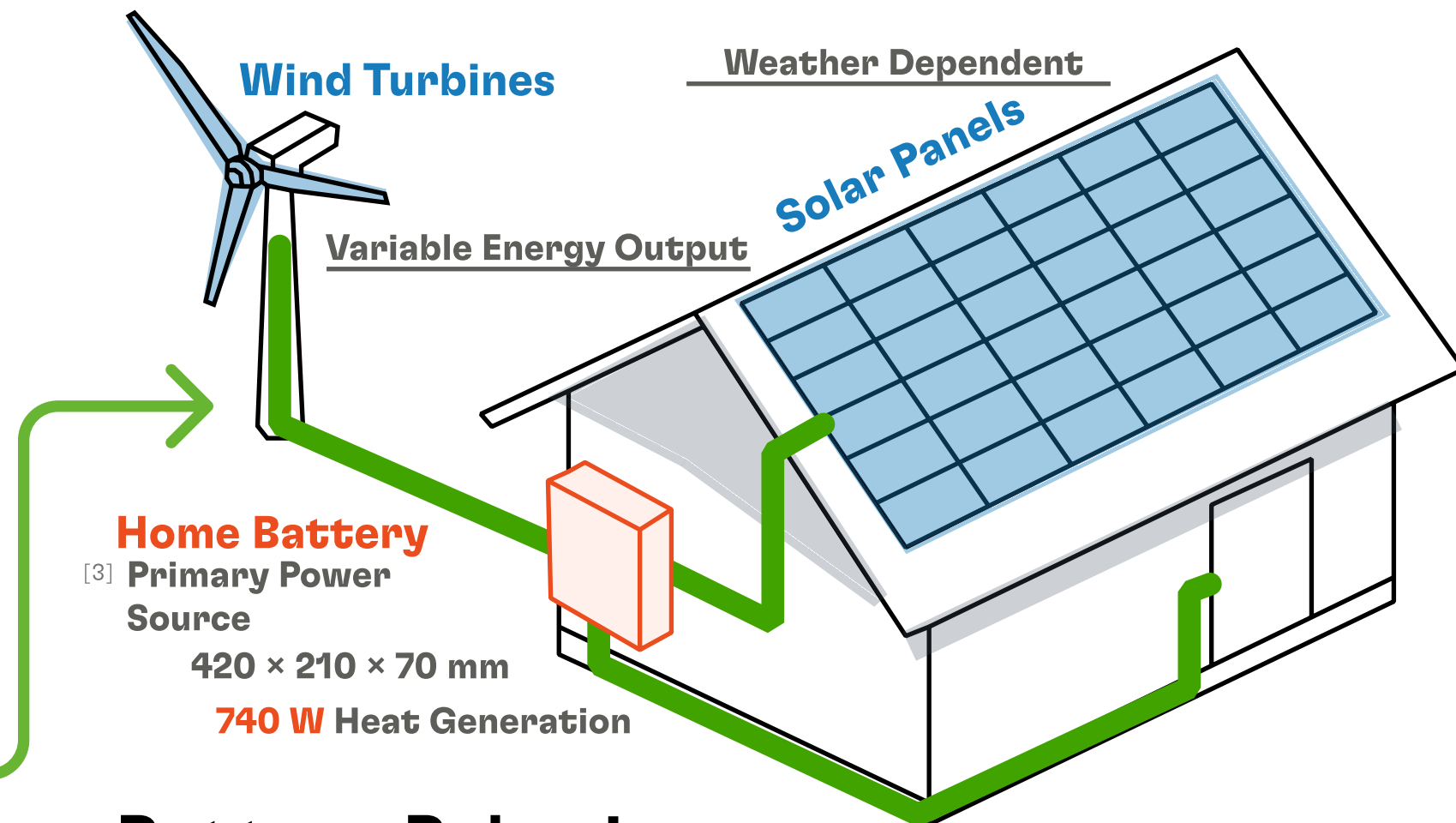
- Any power used by cooling reduces what the user can actually use.
- The system needs to balance removing heat with keeping the energy consumption low.



Cost Effective

Replacing a battery is expensive and difficult in an off-grid system.

- Poor thermal control shortens lifespan, increasing how often replacement is needed.
- Higher pump load and energy use also increase long-term running costs.



Battery Behaviour

The brief specifies a heat generation of **120,000 W/m³** (2C discharge). Using the given size **420 × 210 × 70 mm**, this gives a total heat load of about **740 W** that must be removed continuously. [3]



As temperatures approach and exceed **45 °C**, degradation increases more quickly, this is where side reactions inside lithium-ion cells begin to accelerate significantly, reducing lifespan. [11]

Uniformity

The battery is modelled as a single homogeneous block, but in reality it consists of many cells. Temperature differences can develop across the pack, and differences greater than about 5 °C between cells lead to uneven ageing, where hotter cells degrade faster. **Since individual cells are not modelled, this is instead controlled by limiting the max-average temperature difference across the battery.**

Design Aims

Max battery temperature
<45 °C

Limits accelerated degradation and protects overall lifespan.

Max-average temperature difference
<8 °C

Reduces risk of uneven ageing, and battery capacity loss

Low average temperature
<40 °C

Reduces thermal stress during continuous operation

Low pressure drop
<500 Pa

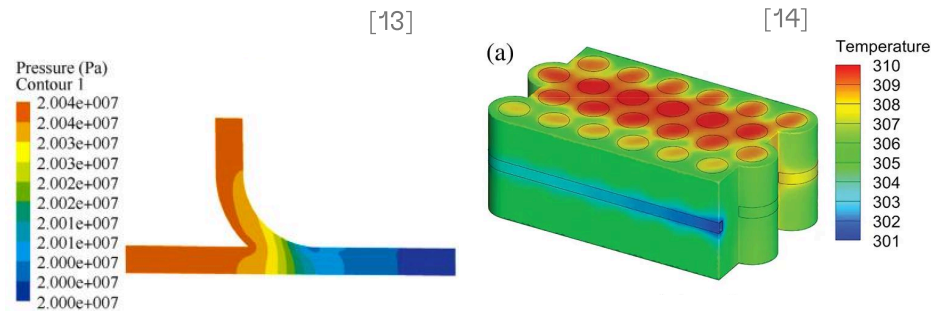
Reduces required pump force, minimising the electricity used for cooling.

Rationale & Development

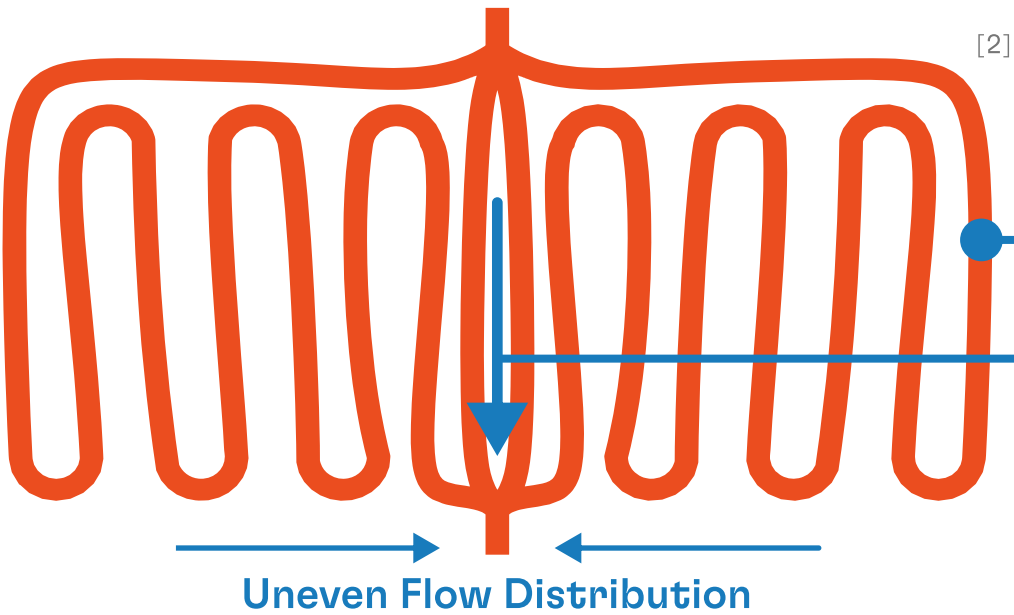
The perfect battery for Green Gary

Initial Rationale

Instead of choosing a target user the aim was to **improve even heat distribution** of the battery focussing on the central hotspot as can be seen below. In order to target this, the initial design focused on **increasing the flow** through the **centre** of the cooling plate with an increased number of shorter channels. **Curved junctions** were also incorporated to create smoother flow transitions, with the aim of **reducing sharp changes in direction** and improving flow distribution through the channel network.



Initial



Issues Identified

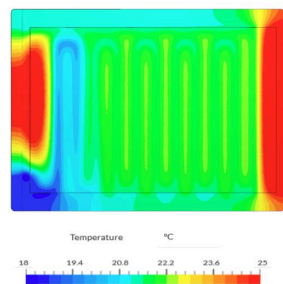
Upon initial inspection we identified the following potential issues with the design:

Less coolant
Significantly Higher Flow

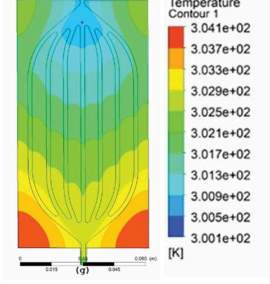
The result of having shorter central channels would lead to flow maldistribution. The distribution of coolant is focused on the central channels and while this was the aim of this design it would lead to a significant decrease in cooling performance in the outer loops.

Benchmarking

Serpentine [15]



Parallel [16]



Serpentine ends up with a large **lack of uniformity** across the plate. Whereas parallel channels lead to more uniformity but have uneven flow distribution.

Hence **combining both** we hoped to **mitigate these issues** in our initial design & create a composite, better design.

Curves

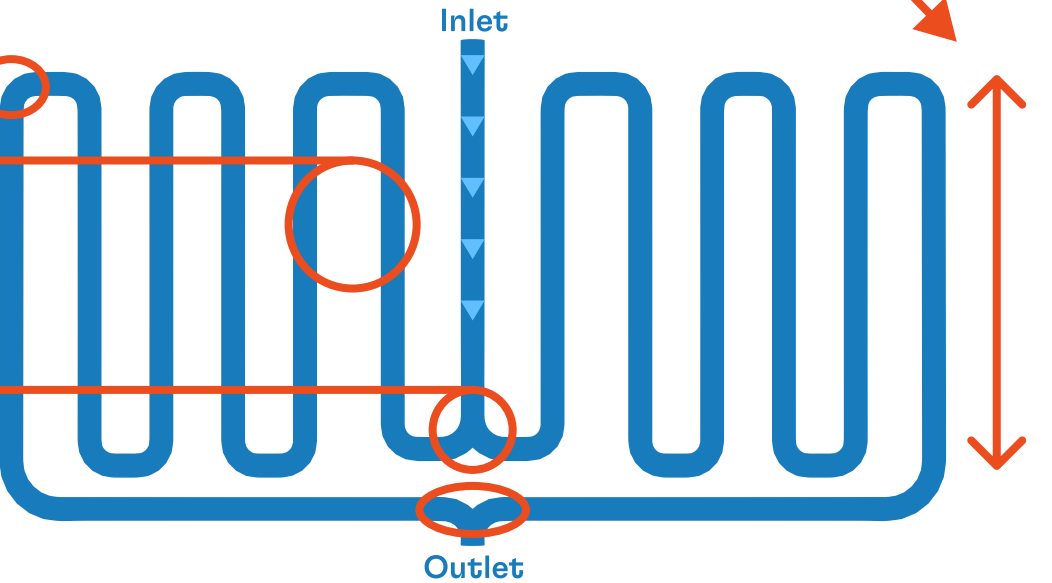
Any changes in channel direction are curved to reduce flow separation.

Geometry Simplified

Straighter pipes creates a reduction in pressure drop & turbulence.

Removed central channel
Eliminates flow bias

Refined



Less channels converge
Removing the central channel means less turbulence & pressure occurs at the outlet.

Constraints

These are fixed modelling conditions from the handbook that define the CFD setup:

- Initial temperature (battery + coolant): 25 °C
- Coolant: water
- Cooling plate material: aluminium 6061-T4
- Inlet: mass flow rate = 12 g/s
- Outlet: pressure = 0 Pa (pressure outlet)
- Contact thermal resistance between battery and plate neglected

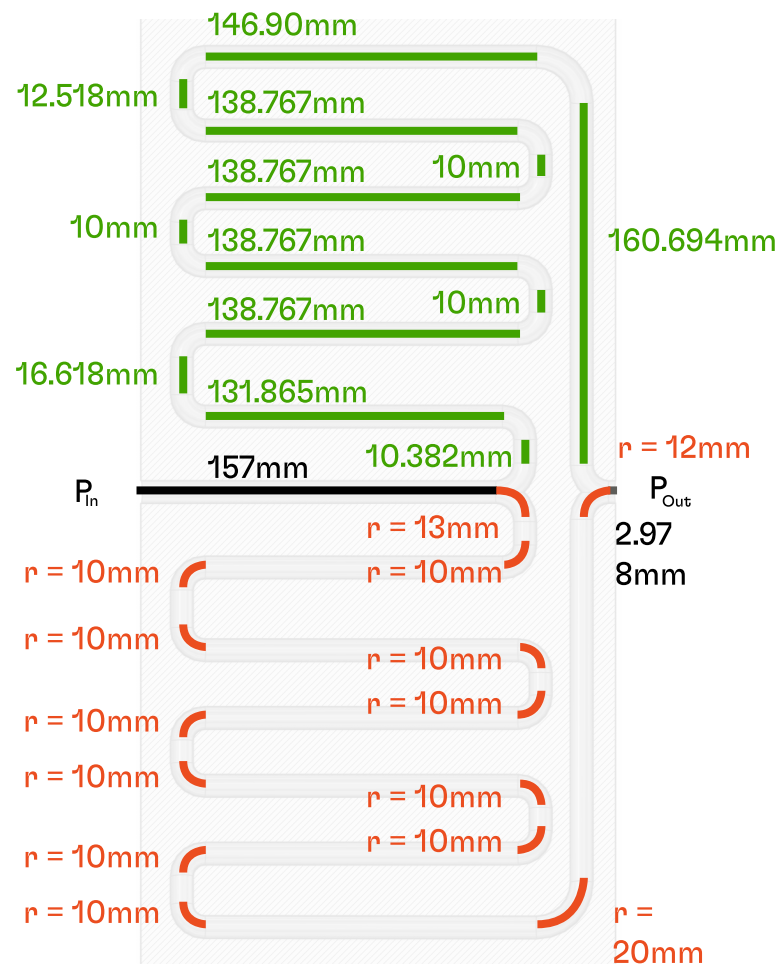
Design Summary

This final developed concept is a **parallel serpentine layout** that aims to cool all cells in the pack evenly instead of just focussing on cooling the hottest areas the most. In addition we **straightened channels** to allow for more even & predictable heat dissipation. We have also featured more **consistent bend angles** and more even channel distribution across the cooling plate.

Manufacturability

Our revised design, considered **ease of manufacture** as part of the design considerations. Assuming this plate is manufactured using a **subtractive manufacturing** method such as CNC milling. Hence using a constant channel diameter of **10mm** allows for **easy machining**, smooth bends prevent the need for more complex tooling keeping manufacturing **cost down** ensuring **market viability**.

Measurements:



ρ	Fluid Density	1000 kg/m ³
$\dot{m}$	Mass Flow Rate	0.012 kg/s
A	Cross-sectional area of pipe	$\pi (10 \cdot 10^{-3})^2 / 4 = 25 \cdot 10^{-6} \cdot \pi \text{ m}^2$
D	Hydraulic Diameter	0.010 m
μ	Dynamic Viscosity	0.001 Pa•s

$$\Delta P_{\text{Total}} = \Delta P_{\text{Combined}} + \Delta P_{\text{Parallel}}$$

$$\Delta P_{\text{Parallel}} = \Delta P_{\text{Straight}} + \Delta P_{\text{Bends}}$$

ΔP
Combined

$$\Delta P_{\text{Straight Pipe}} = f \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2}$$

- f Darcy Friction Factor
- L Total Pipe Length
- D Hydraulic Diameter
- ρ Fluid Density
- v Fluid Flow Velocity
- $\dot{m}$ Mass Flow Rate
- A Cross-sectional area of pipe

$$v = \frac{\dot{m}}{\rho A}$$

Reynold's Number:

$$\text{Re} = \frac{\rho v D}{\mu} = \frac{\dot{m} D}{\mu A}$$

Since we don't have v but have m , we substitute.

$$\text{Re}_{\text{Combined}} = \frac{0.012 \cdot 0.01}{0.001 \cdot 25 \cdot 10^{-6} \cdot \pi}$$

$$= 1527... < 2300 \quad \therefore \text{Laminar}$$

Darcy Friction Factor for Laminar Flow:

$$f = \frac{64}{Re} = \frac{64}{1527} = \frac{\pi}{75}$$

Length of Combined Flow Pipes

$$L = 0.157 + 0.00297 = 0.15997\text{m}$$

$$\Delta P_{\text{Combined}} = f \cdot \frac{L}{D} \cdot \frac{\dot{m}^2}{2\rho A^2}$$

$$= \frac{\pi}{75} \cdot \frac{0.15997 \cdot 0.012^2}{0.01 \cdot 2 \cdot 1000 \cdot (25 \cdot 10^{-6} \cdot \pi)^2}$$

$$= 7.82 \text{ Pa}$$

ΔP
Straight

$$\dot{m} = 0.012 / 2 = 0.006$$

Two identical parallel paths split the mass flow rate in half.

$$Re = \frac{\rho v D}{\mu} = \frac{\dot{m} D}{\mu A}$$

Reynold's Number:

$$\text{Re}_{\text{Parallel}} = \frac{0.006 \cdot 0.01}{0.001 \cdot 25 \cdot 10^{-6} \cdot \pi}$$

$$= 763.. < 2300 \quad \therefore \text{Laminar}$$

Darcy Friction Factor for Laminar Flow:

$$f = \frac{64}{Re} = \frac{64}{763...} = \frac{2\pi}{75}$$

Length of straight pipe in a single branch of pipe system:

$$L = (10.382 + 10 \cdot 3 + 138.767 \cdot 4 + 131.865 + 16.618 + 12.518 + 146.9 + 160.694) \cdot 10^{-3} = 1.06405 \text{ m}$$

$$\Delta P_{\text{Straight}} = f \cdot \frac{L}{D} \cdot \frac{\dot{m}^2}{2\rho A^2}$$

$$= \frac{2\pi}{75} \cdot \frac{1.064045 \cdot 0.006^2}{0.01 \cdot 2 \cdot 1000 \cdot (25 \cdot 10^{-6} \cdot \pi)^2}$$

$$= 26.01 \text{ Pa}$$

$$\Delta P_{\text{Combined}} + \Delta P_{\text{Straight}} + \Delta P_{\text{Bends}} = 111.5 \text{ Pa}$$

ΔP
Bends

For bends we use the minor loss equation:

$$\Delta P_{\text{Bend}} = K \cdot \frac{\rho v^2}{2} = K \cdot \frac{\dot{m}^2}{2\rho A^2}$$

K Loss coefficient

K is dependent on R/r (Pipe Radius/Bend Radius) determined by empirical data. However most data available for K is for turbulent flow with high Reynold's number. As we have low Reynold's we need to use:

Hooper 2K Method

$$K = \frac{K_1}{Re} + K_{\infty} \cdot \left(1 + \frac{1}{D_{in}}\right)$$

D_{in} Pipe diameter in inches

R/r	K ₁	K _∞	K
1.0	800	0.25	1.94
1.5	800	0.2	1.76

$$\Delta P_{\text{Bend}} = \Sigma K \cdot \frac{m^2}{2\rho A^2}$$

$$R/r = [11 \cdot 1.0, 3 \cdot 1.5]$$

$$\Sigma K = (11 \cdot 1.94 + 3 \cdot 1.76) = 26.62$$

$$\Delta P_{\text{Bend}} = 26.62 \cdot \frac{0.006^2}{2 \cdot 1000 \cdot (25 \cdot 10^{-6} \cdot \pi)^2}$$

$$= 77.67 \text{ Pa}$$

As Hooper 2-K's laminar term predicts higher values than reality for low Re numbers. Therefore this should be an overestimate.

1D Heat Transfer

Hand Calculations ^[17]

Given Data

Battery Pack Dimensions	L = 0.42 m, W = 0.21 m, H = 0.07 m
Battery Conductivity	K _b = 30 W/m•K
Volumetric Heat Generation	q''' = 120000 W/m ³
Ambient	T _∞ = 25 C, h _∞ = 5 W/m ² •K
Cold Plate Conductivity	K _p = 170 W/m•K
Tube diameter	D = 0.010 m
Coolant	Water, T = 20 C, ρ = 997 kg/m ³ , μ = 0.00089 kg/m•s, c _p = 4180 J/kgC, K = 0.6 W/m•K
Mass Flow Rate	ṁ = 0.012 kg/s

Conservation of Energy

$$Q = q''' \cdot V = 120000 \cdot 0.42 \cdot 0.21 \cdot 0.07 = 740 \text{ W}$$

Conservation of energy tells us this: $Q_{\text{In}} = Q_{\text{Out}}$

- 1 Convection to ambient via 5 exposed faces
- 2 Conduction through the cold plate (bottom face)
- 3 Plus, convection through the pipe/coolant liquid
- 4 Plus, convection through the bottom surface of the plate to surrounding environment/ambient

We can expect the thermal resistance for convection to the surroundings to be significantly larger than compared to the cold plate, so we assume all 637 W is transferred to the cold plate.

Flow Regime

First we need to calculate Reynold's number:

$$Re = \frac{\rho v D}{\mu} = \frac{m D}{\mu A}$$

$$Re = \frac{0.012 \cdot 0.01}{0.001 \cdot 25 \cdot 10^{-6} \cdot \pi} = 1527... < 2300$$

∴ Fully Developed Laminar Flow

The Nusselt Number can be assumed based off of the flow regime, however we have to take into account that there can be localised turbulence in the bends.

For fully developed laminar flow under constant heat flux:

$$Nu = 4.36$$

However, Nu can be much higher for turbulent flow so we use an enhancement factor:

$$Nu_{\text{eff}} = Nu \cdot 1.8 = 7.848$$

Thermal Resistance

Due to Convection:

$$R_h = \frac{1}{h A_{\text{wetted}}}$$

h, heat transfer coefficient
A, area in contact with coolant.

$$A_{\text{wetted}} = \pi D L_{\text{total}} \quad L_{\text{total}} = 2.782 \text{ m}$$

$$A_{\text{wetted}} = 0.087 \text{ m}^2$$

Heat transfer coefficient:

$$h = \frac{Nu K}{D} = \frac{7.848 \cdot 0.6}{0.010} = 470 \text{ W/m}^2 \text{ K}$$

$$R_h = \frac{1}{470 \cdot 0.087} = 0.0244 \text{ K/W}$$

Thermal resistance due to Convection:

$$R_{\text{cond}} = \frac{L}{K A_{\text{contacted with battery pack}}}$$

L, cold plate thickness

K, thermal conductivity of cold plate

$$R_{\text{cond}} = \frac{0.003}{170 \cdot 0.088} = 0.0002 \text{ K/W}$$

Therefore now we can calculate total thermal resistance as:

$$R_{\text{Tot}} = 0.0246 \text{ K/W}$$

$$\Delta T = Q \cdot R_{\text{Tot}} = 18.2 \text{ C}$$

If we can find the average temperature of the coolant we can then subsequently estimate the surface temperature of the pack.

$$T_{\text{Avg}} = \frac{T_{\text{Out}} + T_{\text{In}}}{2} \quad T_{\text{In}} = 20 \text{ C}$$

$$\Delta T = \frac{Q}{\dot{m} c_p} = \frac{740}{0.012 \cdot 4180} = 14.75 \text{ C}$$

$$T_{\text{Avg}} = \frac{\Delta T + 2 T_{\text{In}}}{2} = 27.375 \text{ C}$$

Battery Temp

Since we know the difference between the surface temperature of the battery pack and the average temperature we know the surface temperature to be:

$$T_s = 27.375 + 18.2 = 45.575 \text{ C}$$

$$R_{\text{cond battery}} = \frac{L}{K A_c}$$

L, height of the middle of the battery pack
K, thermal conductivity of battery

$$R_{\text{cond battery}} = 0.013 \text{ K/W}$$

$$\Delta T = Q \cdot R_{\text{cond battery}}$$

$$= 740 \cdot 0.013 = 9.62 \text{ C}$$

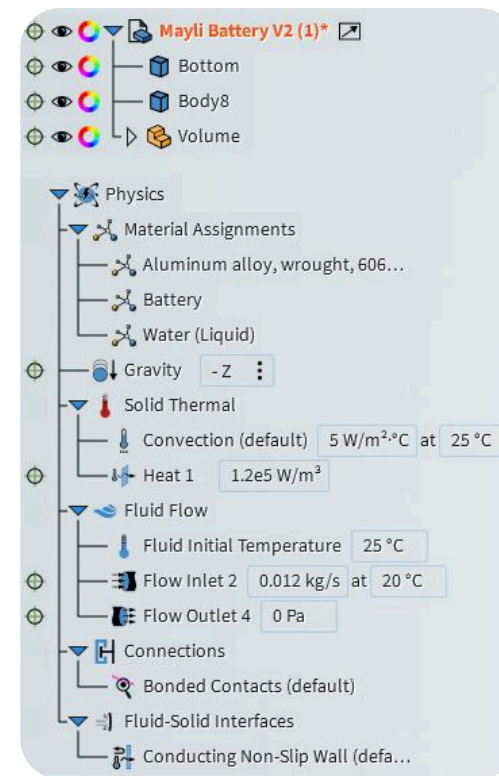
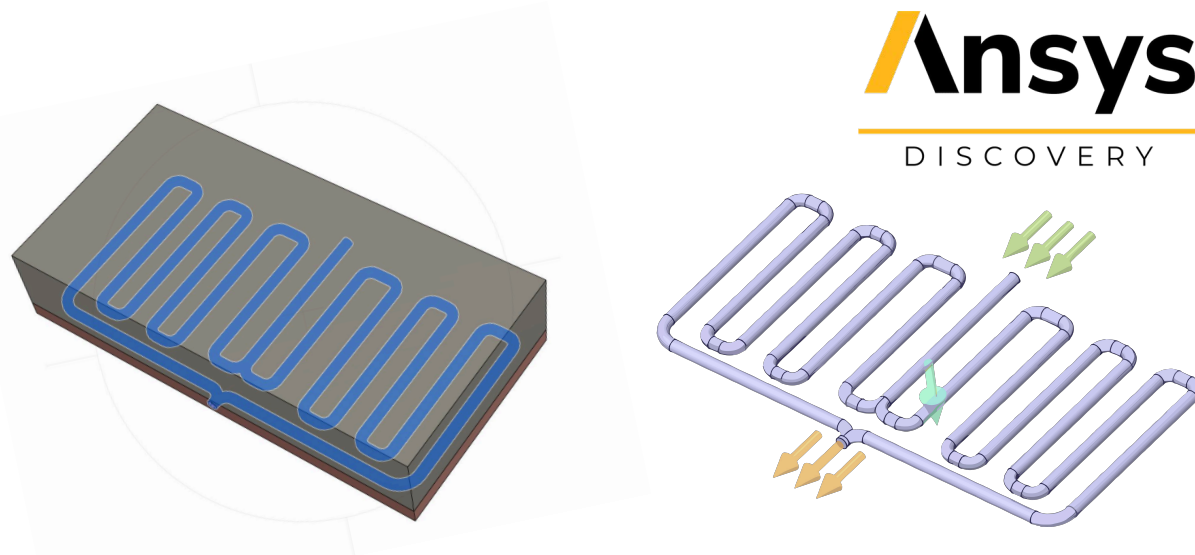
Now we know the difference in temperature between the middle and the bottle of the battery pack. We can estimate the average temperature.

$$T_{\text{Avg}} = \Delta T + T_s = 55.1 \text{ C}$$

This value **will be different** from our CFD as we have **arbitrarily determined** the enhancement factor of Nu. Without **knowing how turbulent** the coolant is around our design we cannot know for certain what Nusselt Number to use and therefore **cannot use** this as a **reliable way** to predict the **CFD values**. Our prediction is due to the **low turbulence** we should get in our pipes Nu is actually an **overestimate** and so is this average temperature.

CFD Setup

How do we test it?



Limitations of Process

Use of Discovery

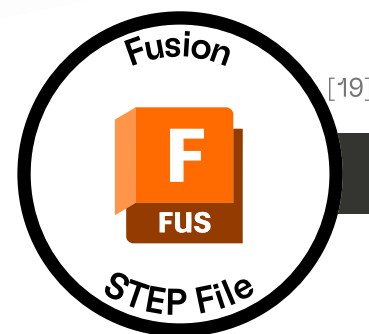
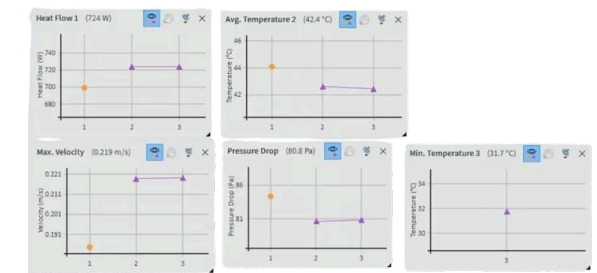
Ansys discovery is a redesigned, **abstract** interface for Ansys mechanical & fluent. This means as the user we lose a **large** amount of **control** over the **settings** used at each step in the process. This means that we can **lose** accuracy in our results as we cannot **refine** the solution setup as much as we did in **report 1-1**.

Student License

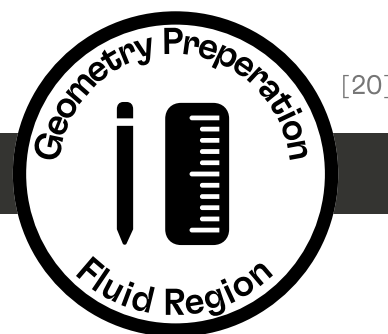
The **student** license significantly limits the **cell count** of simulations to **512k**. This means that the mesh is **coarser** and **less accurate** results can be **derived**.

Modelling Assumptions

Discovery limits the **number of models** we can **choose from** meaning we can't use the **best fitting options**.



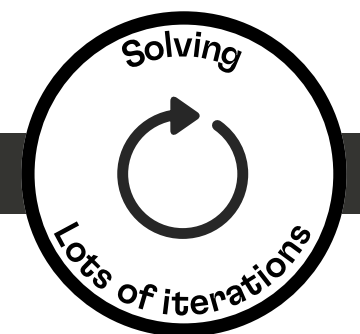
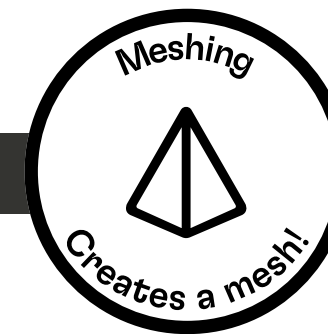
[19]



[20]



[3]



[3]

Fusion

The desired **geometry** was created in **Fusion 360**, including the **battery**, **plate**, and **channel** cut-outs. The **channel** cut-outs directly **matched** the **design** used in the **hand calculations**.

Geometry Preparation

Once we had the **.Step** file we were able to import it into **Ansys Discovery** and begin the setup process. The first step was to **create** the **volume** that represented the **fluid** region within the **cooling** plate aka the **volume** for the **cooling channel**.

Solution Setup

Next we setup the **simulation** with all it's **boundary & environmental conditions**. These include the **materials** for each volume, the **Fluid Flow** Parameters & **solid thermal** conditions.

Meshing

Now the solution is set up as specified we need to **create** a mesh that can be used by **refine** mode, a **high fidelity** **CPU** based solver that is **able** to calculate **key measures** about our **specified** model.

Solving

Once the **simulation** has been setup we run it using the **Refine** mode as part of **discovery**. We also **define** any **variables** we want to measure and calculate as part of the simulation before we run it, these are called **monitors**.

Process

1. A **sketch** was created using an image of the design.
2. Fusion's **pipe** tool was used on the **sketch** to make a solid pipe of diameter 10mm.
3. A plate with dimensions **420 × 210 × 16 mm** was created, and the pipe was cut out of it.
4. The battery was then created on top of the plate with dimensions **420 × 210 × 70 mm**.

Issues Encountered

With **multiple** pipe networks when there were **multiple** inlets and outlets this had to be done **differently**.

Process

1. Select the **Volume** Extract Tool
2. Select the **edge** loop of the **inlet(s)**
3. Select the **edge** loop of the **outlet(s)**
4. Complete the **operation**

Process

1. Select the **fluid flow** tool and define the **inlet(s)** to have a constant flow rate of **0.012 kg/s**
2. Define the **outlet(s)** to have a pressure of **0 Pa**.
3. Ensure the **fluid** region is defined as **water** & the plate is set to its relevant **material**.
4. Select the **battery** volume and define it as a **1.2e5 W/s** **heat** source with its defined **material**.

Process

1. Select the **mesh fidelity** level. We found the **lowest** worked best to ensure the **mesh** is **not too** complex to **avoid** license **limitations**.
2. Inspect the **mesh preview**
3. Add **local sizing** if needed to areas of **higher** fidelity E.g. Inlet
4. **Check** mesh quality measures using **inbuilt** Ansys tools.

Element Count: **777961**

Process

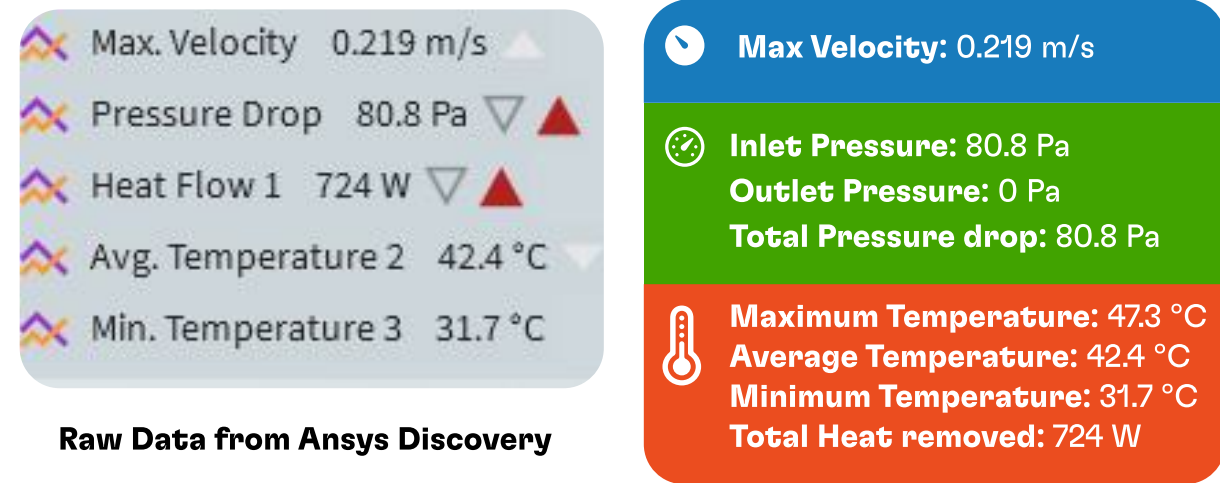
1. Define **Monitors**, E.g. Average battery temp.
2. Run **Simulation**

Issues Encountered

When the mesh has a **fidelity** that is **too high** the **simulation** can fail due to **licensing** issues. When this **occurs** we **reverted** back to the **meshing** step and increased the **cell size** to attempt to **reduce** the **total cell count**.

CFD Outputs

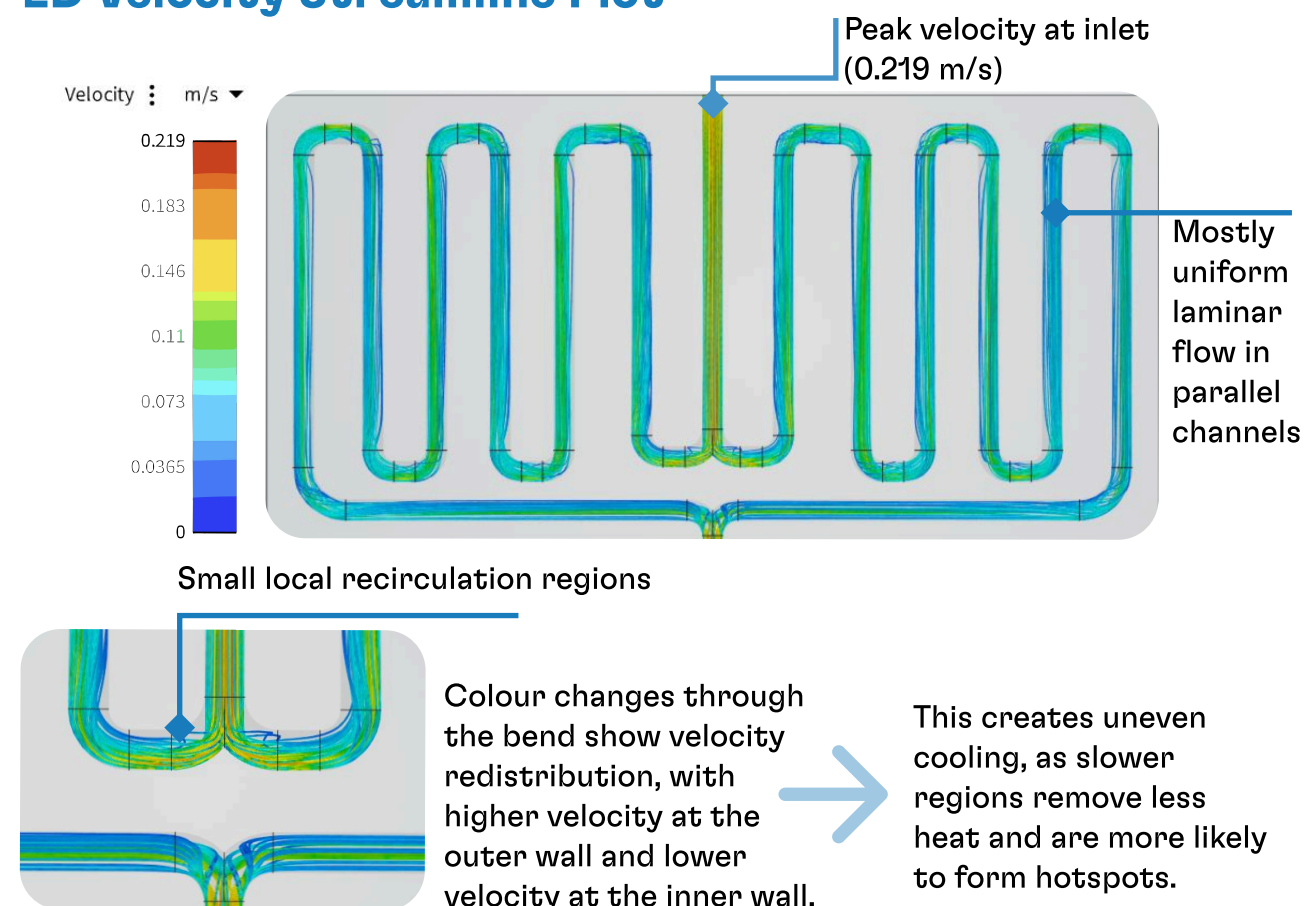
Not so pretty pictures



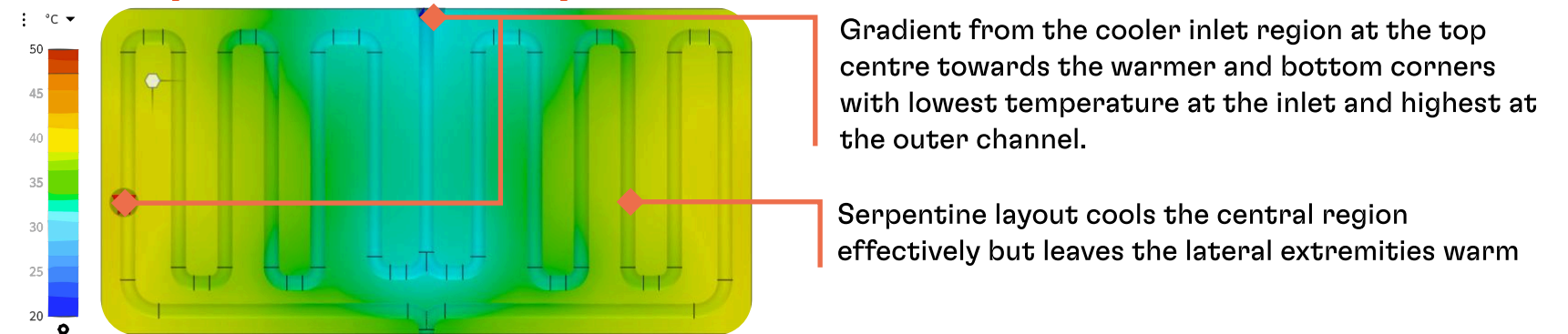
Raw Data from Ansys Discovery

CFD outputs are **promising** but don't meet all the targets. The maximum temperature of **47.3 °C** and average temperature of **42.4 °C** are both above the desired limits. The temperature spread of **4.9 °C** is within the **<10 °C** uniformity target, showing fairly even heat distribution. The pressure drop of **80.8 Pa** is well below the **<500 Pa** limit. Overall, pressure drop and thermal uniformity are good, but further optimisation could be done and the battery temperature needs to be reduced.

2D Velocity Streamline Plot

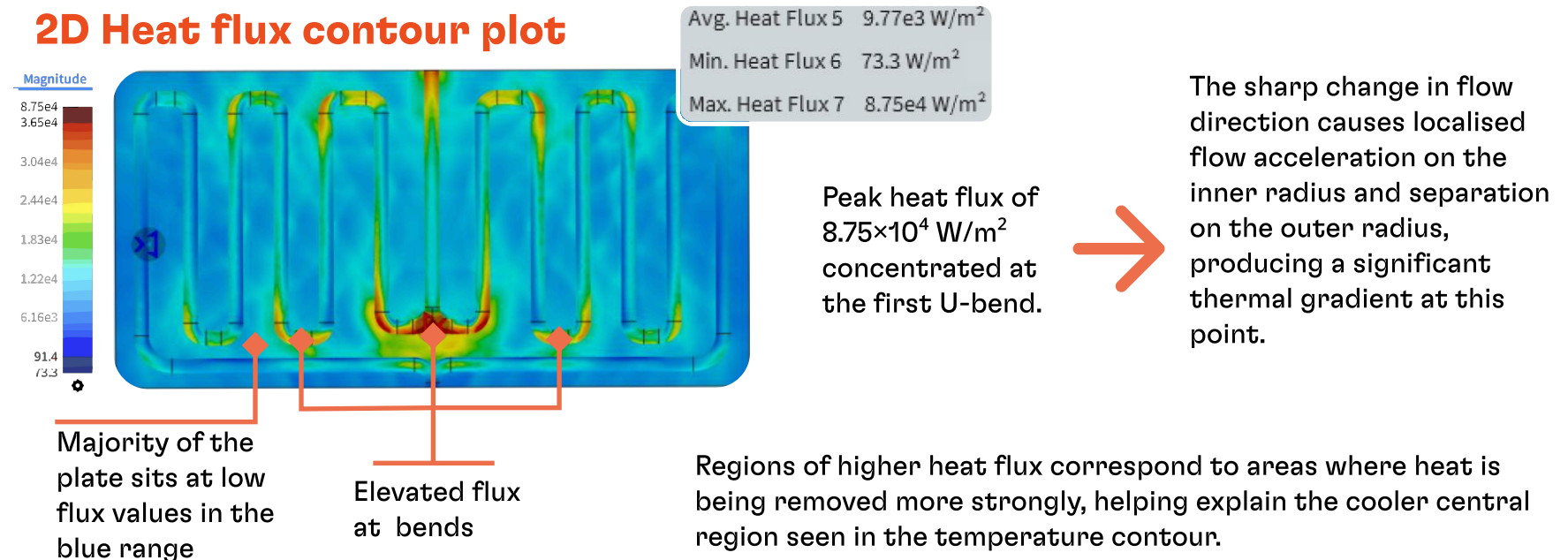


2D Temperature contour plot

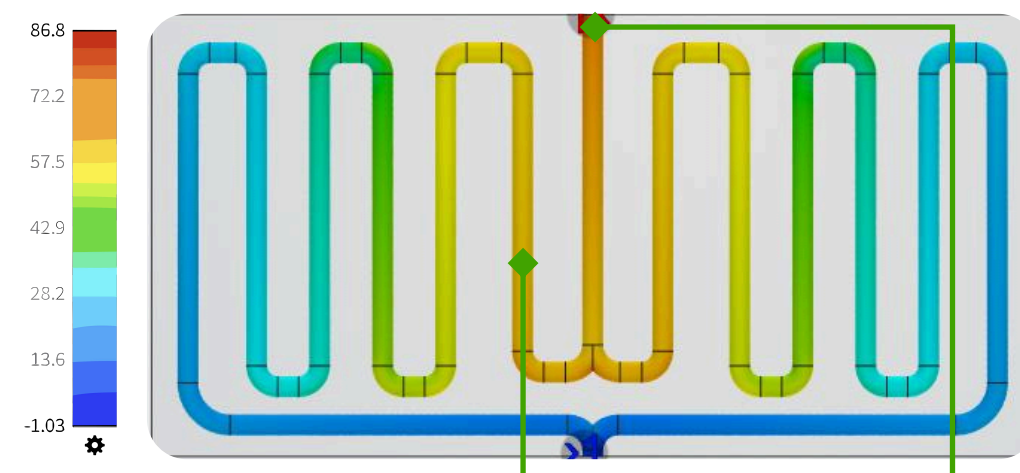


The temperature contour shows where the plate is hotter or cooler overall, while the heat flux contour shows where heat is being removed most strongly

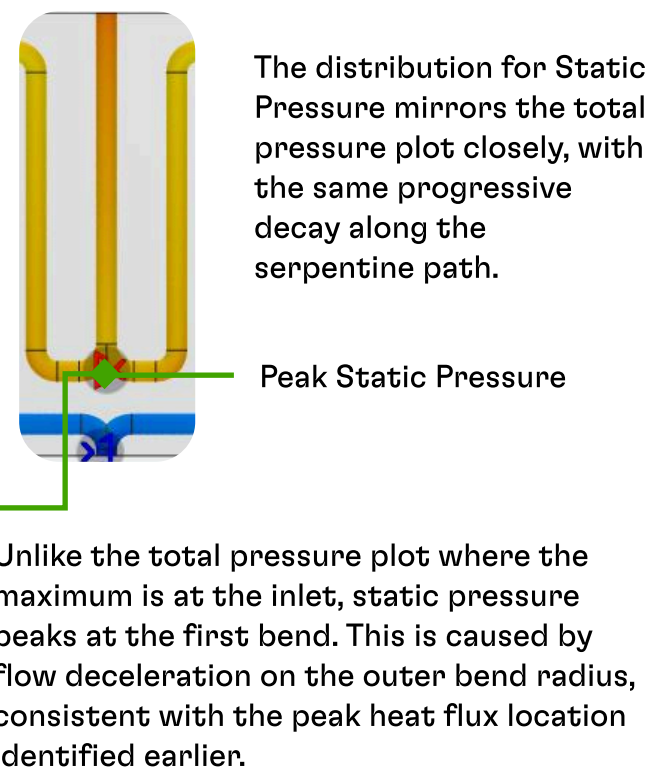
2D Heat flux contour plot



Total Pressure Plot



Static Pressure Plot



Hand vs CFD

Agreements and Discrepancies

Optimisation Technique

Notably we **did not use** our **hand calculations** as a method of **optimising** the design for pressure loss or heat transfer. This is for **several reasons**: **firstly**, the **models** we used to calculate pressure loss and heat transfer by hand were **extremely simple** with **many assumptions** that **limited** the **accuracy** of the **prediction**; **secondly**, **values** in both calculations **were assumptions** that would need to be **validated** by **empirical data** or **CFD** to **justify** their value. Therefore without a **reliable result** or **method**, it is **unsuitable** for use as an **optimisation technique**.

Pressure Loss 

Hand  111.5 Pa

CFD  80.8 Pa

ERROR⁴⁰⁴ 30.7 Pa

Minor Loss Equation

$$\Delta P_{\text{Bend}} = K \cdot \frac{\rho v^2}{2} = K \cdot \frac{\dot{m}^2}{2 \rho A^2}$$

The **minor loss equation** is used for determining the **pressure loss** in pipe **fittings** such as **elbows** and **forks**. It relies on something called a **loss coefficient**. Loss coefficient is found from empirical data.

This **empirical data** is hard to come by for the **specific conditions** of our cooling system as you need to find **loss coefficient** values for your **specific flow** (turbulent, laminar or transitional) with a

R/r (bend radius divided by pipe internal radius) for a flanged elbow of 90 degrees.

As you can imagine, this is **very hard** to find **specific data** for, as most data for pipe flow is for **larger** more **industrial pipe systems** with **turbulent flow** rather than small coolant systems.

Therefore we'd need a **new method** for determining loss coefficients, a method which would be suitable for low Reynold's values:

Re = 763...

And would have **available data** for bends of a **ratio** of **1.0, 1.2, 1.3 and 2.0**. This method was the **Hooper 2-K method**.

Hooper 2-K Method

$$K = \frac{K_1}{\text{Re}} + K_\infty \cdot \left(1 + \frac{1}{D_{\text{in}}}\right)$$

The Hooper 2-K Method is a method of estimating minor losses in a pipe by separating the coefficient into 2:

K_1 the low-Reynolds (viscous) effects

K_∞ the high-Reynolds (turbulent) effects

By finding data for K_1 and K_∞ for a 90 degree flanged elbow we would be able to

calculate a **suitable loss coefficient** for our **low Reynold's** value laminar flow.

This **isn't perfect** and has its **errors** (discussed in **discrepancy**) but by using available data for **K_1** and **K_∞** for **R/r ratios** of 1.0 and 1.5, we can roughly **approximate** the **pressure loss** of the cooling system.

Discrepancy

The discrepancy is likely due to the fact that at **low Reynold's values** Hooper 2-K becomes **strongly viscous** and **geometry dependant** and will greatly **overestimate** the **pressure loss**.

Equivalent Length

An **alternative** method we could try is the **equivalent length** method. Where you **convert bends** to their **equivalent length** which also requires **empirical data**, but often more **accessible**. Then you treat it as though it were a **straight pipe** with an **equivalent length** equal to your calculation.

Heat Transfer 

Hand  55.1 °C

CFD  42.4 °C

ERROR⁴⁰⁴ 12.7 °C

Method Commentary

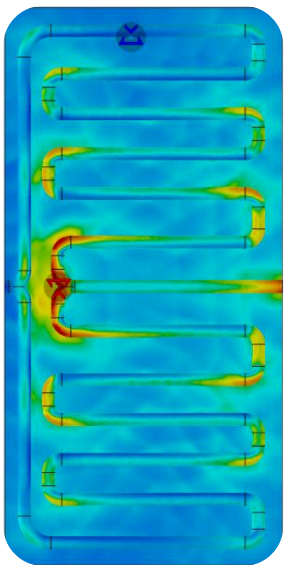
The methodology we used is a **okay estimate** for heat transfer and captures **key physics** in a simple thermal resistance model. Despite being fast, its inaccurate as it's **limited** by its many assumptions (such as **1D heat flow**, **uniform properties**, **estimated Nusselt correction**, **neglecting ambient losses**).

Complex effects like 3D conduction and local turbulence are simplified down to **nothing**. As a result, it provides an correct magnitude value, but detailed methods (e.g. CFD or experiments) are needed for precise predictions.

1 Dimension Limits

For the sake of **simplicity**, we used a **single dimension** of heat transfer. Assuming all heat travels down through the battery **into the plate** and **out the bottom**.

It's simple but **wildly oversimplifies** the **system**. It boils down the cooling plate to essentially an **area** and a **flow rate** rather than a **distribution** of **pipes** throughout a **2D plate**.



Discrepancy

The discrepancy would seem quite high for heat transfer and **that makes sense**, we **expect to see a high difference** because we have **arbitrarily picked Nusselt number**. In this case we have picked 1.8 as it is the value recommended in the tutorial. However, we have **no assurance** this is a correct value - and in fact it assumes fully developed laminar flow, which is unlikely.

Shah Corellation and Graetz [21]

A **better method** would be to use the **Shah (1978) correlation** to calculate a **local Nusselt number** from the **Graetz number** and then **convert this** to a heat transfer coefficient, giving a more **realistic prediction** of **cooling plate performance** than assuming fully **developed laminar flow**.

$$\text{Nu} = 1.953 \text{Gz}^{1/3} \quad \text{Gz} = \frac{D \text{RePr}}{L}$$

Therefore:

$$\text{Nu} = 1.953 \cdot \left(\frac{0.01 \cdot 763 \cdot \sim 6}{\sim 0.087} \right) = 15.8$$

$T_{\text{Avg}} = 46.1 \text{ C}$ **ERROR = 3.7 C**

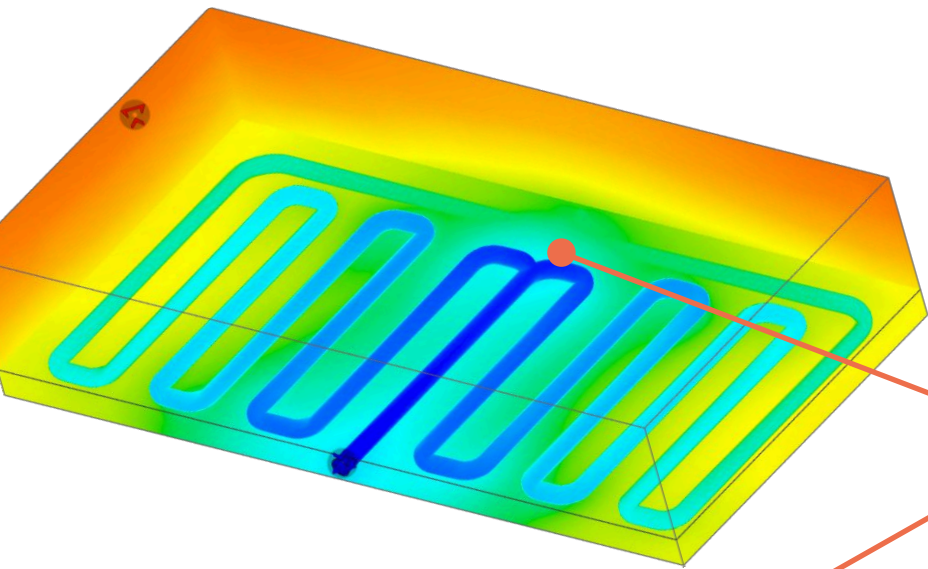
Discussion & Improvements

Making things better (Hopefully)

Discussion of Performance

The CFD results show that the design provides a good balance between cooling performance and hydraulic efficiency, although the design aims were not all met.

3D Temperature Plot



Average Temp: 42.4 °C

Temp Difference = 4.9 °C

Max Temp: 47.3 °C

Aims

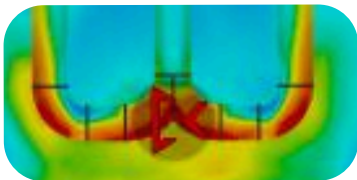
- < 40 °C
- < 8 °C
- < 45 °C

Max and average temperature is still above the aim, showing that further improvement is needed though heat distribution is reasonably uniform and within goals.

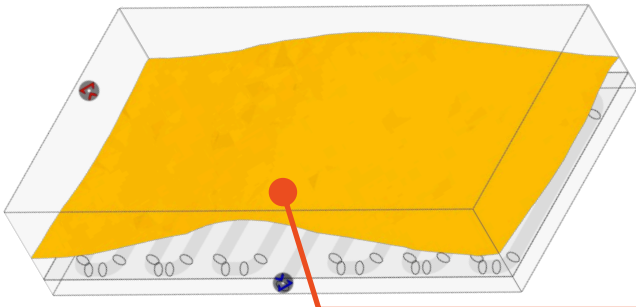
The temperature plot indicates that the central area is cooled more effectively than the outer edges.

The heat flux plot supports this. Heat removal is strongest around the central bend and other bends.

This means the design creates local areas of stronger cooling, so the overall cooling effect is not fully uniform across the battery.

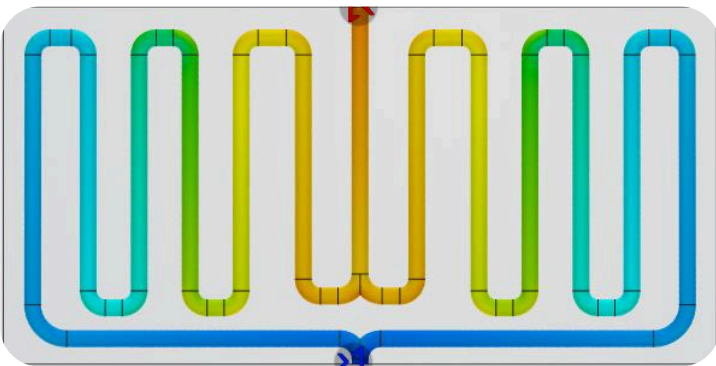


Temperature iso-surface plot



This plot shows the variation in average temperature across the plate. For future improvements, the aim is to reduce this variability and achieve a more uniform thermal distribution.

2D Pressure Plot



Pressure Drop: 80.8 Pa

The total and static pressure plots both show a progressive pressure drop along the serpentine path, consistent with friction and bend losses, but the overall pressure drop is well below the limit we originally stated.

Overall, the main trade-off is that the design already achieves low pressure loss and reasonable thermal uniformity, but the absolute battery temperatures remain slightly too high. This means that further optimisation should focus on improving cooling effectiveness without significantly increasing pressure drop.

Improved Design

The new design focuses on improving temperature uniformity and reducing peak and average temperatures, while also lowering pressure drop.

Multi-inlet, multi-outlet design

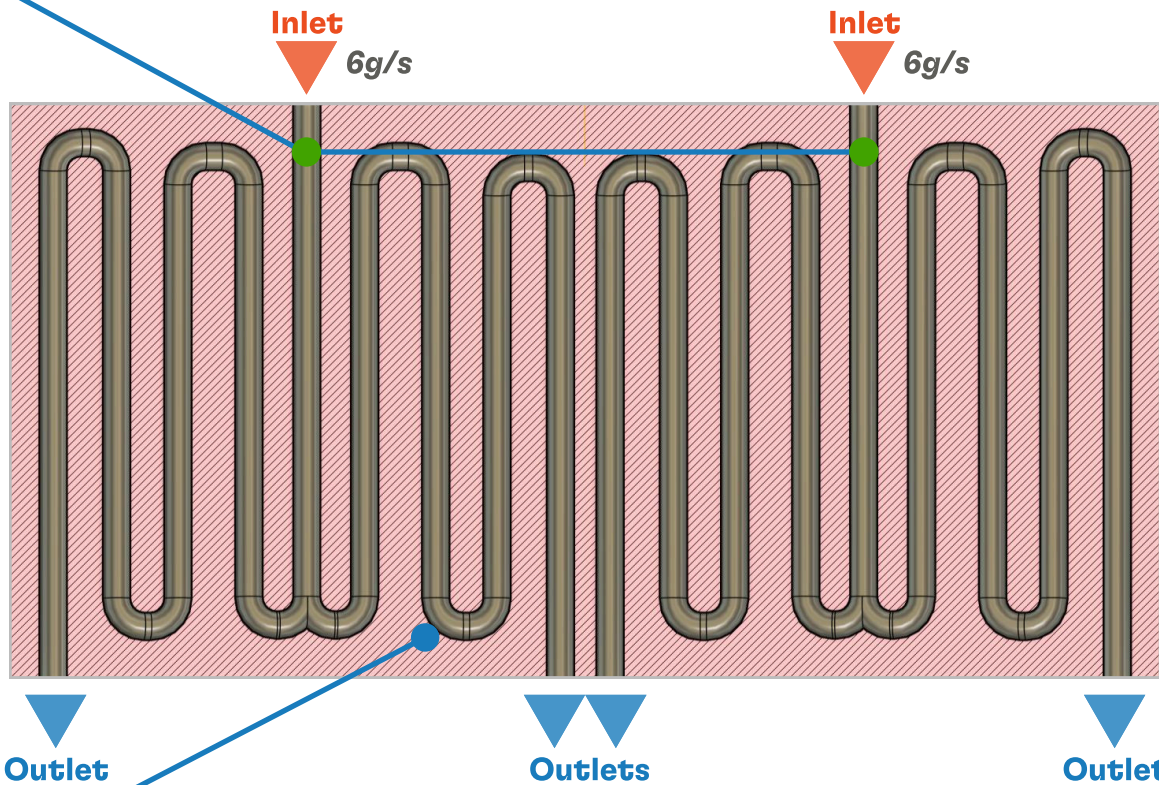
- Splits the flow so coolant does not heat up as much along each path
- Shorter flow paths reduce temperature rise through the channel
- Leads to more even cooling across the plate

10mm Diameter

2 Inlets

4 Outlets

12 Total Bends



Improved coolant distribution across the plate

- Original design cooled the centre more effectively than the edges
- New layout spreads flow more evenly across the full area
- Reduces edge hot spots and improves overall uniformity

Expected Improvements

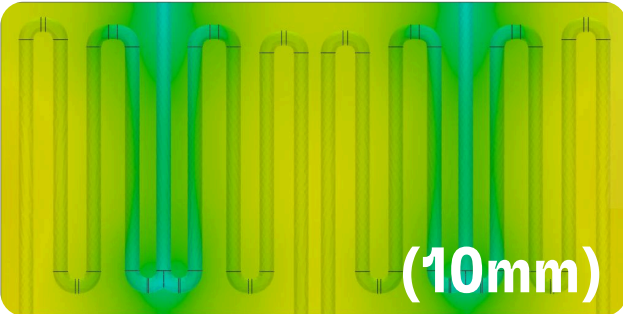
The revised design is expected to reduce peak temperature, improve temperature uniformity, and further reduce pressure drop.

CFD Iteration

Too much CFD

We ran a series of CFD iterations across five design variables to find the optimal cooling plate configuration.

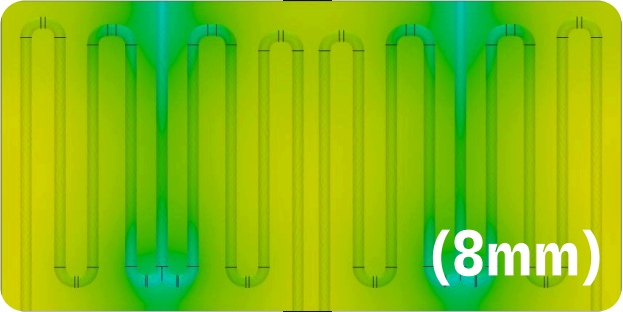
Baseline



Stats | Average Temp: 43.9°C
Max Temp: 46.9°C
Pressure Drop: 19.9Pa

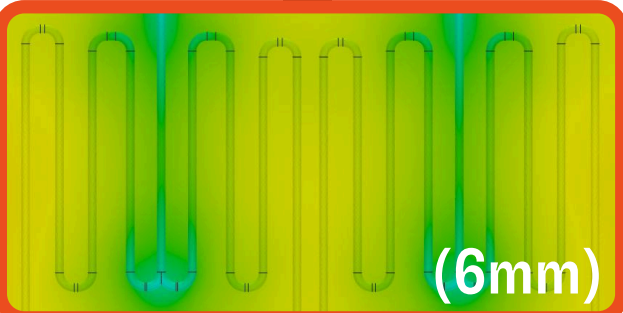
Initially we started with our proposed redesign, while this increased average temp it improved upon temperature uniformity (3°C)

Diameter Optimisation



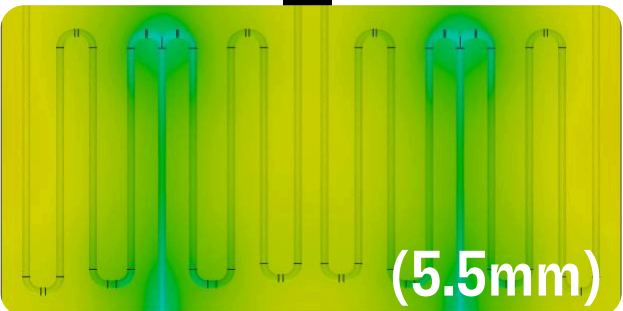
Stats | Average Temp: 43.3°C
Max Temp: 46.1°C
Pressure Drop: 47.1Pa

Next we tried decreasing channel diameter aiming to decrease average temp. This caused an increase in pressure drop.



Stats | Average Temp: 43.9°C
Max Temp: 46.9°C
Pressure Drop: 155Pa

Decreasing the diameter increased pressure drop & lead to worse thermal performance however this allows for geometric benefits.

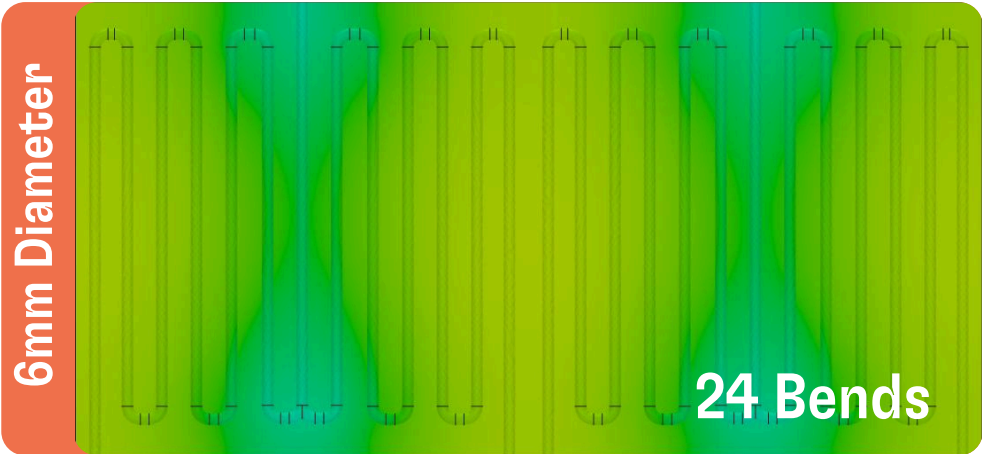


Stats | Average Temp: 43.9°C
Max Temp: 46.9°C
Pressure Drop: 222Pa

We found that at 5.5mm there was no noticeable benefit to decreasing the diameter. In this case the increased pressure drop has no benefit to cooling or geometric capability.

Geometry Changes

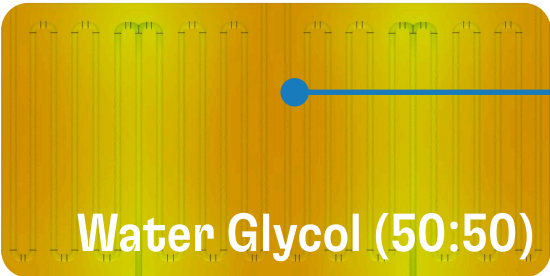
Based off of the results from the diameter optimisation we settled on a diameter of 6mm as it offers comparable cooling performance to our baseline while being narrower allowing us to increase the number of parallel channels within the cooling plate



Stats | Average Temp: 42.6°C
Max Temp: 45.6°C
Pressure Drop: 299Pa

By increasing the number of bends 12 to 24 improves horizontal temperature uniformity. This is due to the fact that at the bends there is significantly more turbulence which increases pressure but is able to remove more heat.

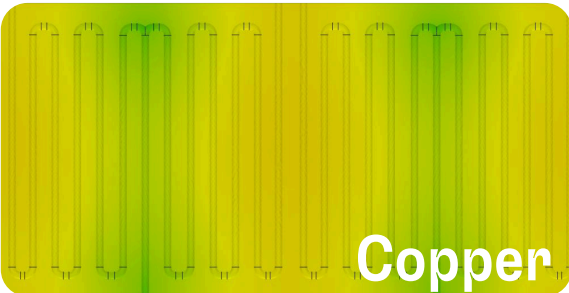
Environmental Changes



More orange profile meaning higher overall temperatures

Stats | Average Temp: 48.2°C
Max Temp: 51.3°C
Pressure Drop: 683Pa

Put simply water glycol is a suboptimal option for home cooling solutions where the temperature of the coolant would rarely lead to it freezing. It drastically impacts the thermal performance and increases the pressure drop.



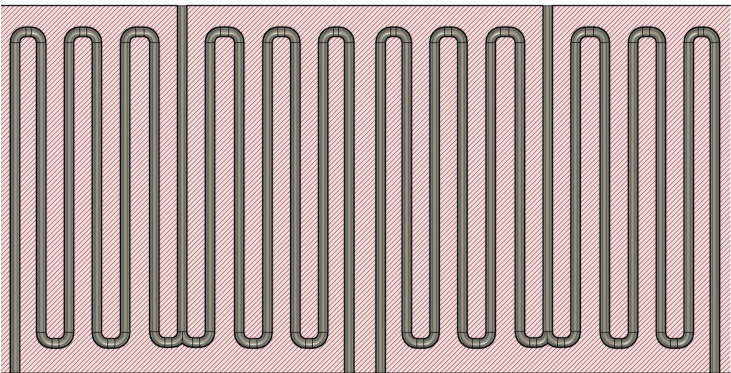
Stats | Average Temp: 42.6°C
Max Temp: 45.6°C
Pressure Drop: 299Pa

Changing the cooling plate material to copper has quite literally no impact on the thermal performance of the plate therefore making the additional economic cost pointless.

Universal Scale
Temperature °C



Optimised Design



Cool Changes
Diameter = 6mm
2 Inlets
4 Outlets
24 Bends
Plate Thickness = 10mm
Overall improved temperature uniformity.

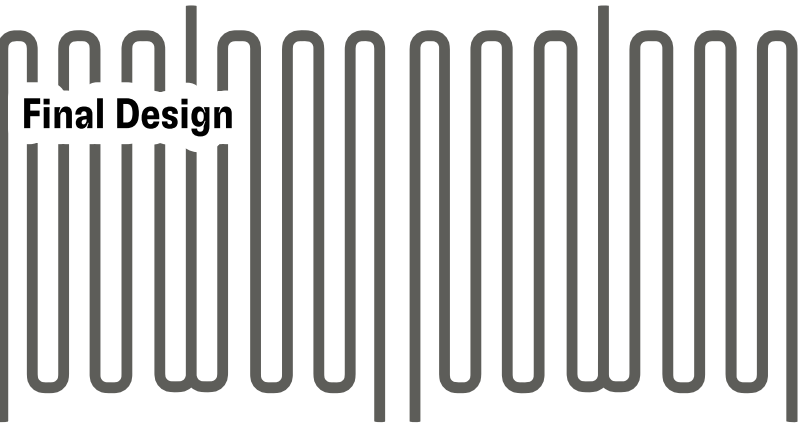
Overall we have carried out a series of optimisations across 5 different variables to find an optimal combination of parameters. This optimised version is a modification of the final geometry change above where we have decreased the plate thickness to 10mm to decrease the distance between the fluid region and the battery as suggested by hand calcs.

Thermal resistance due to Convection:

$$R_{\text{cond}} = \frac{L}{KA} \text{ contacted with battery pack}$$

Final Analysis

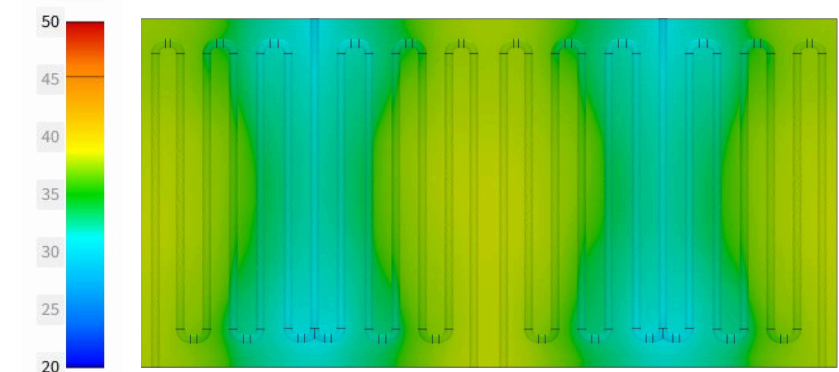
We really really enjoyed doing this



Statistics

Average Temp: 42 °C
Pressure Drop: 297Pa
Max Temp: 45.2°C
Min Temp: 33.4°C
 $\Delta T = 11.8\text{ °C}$

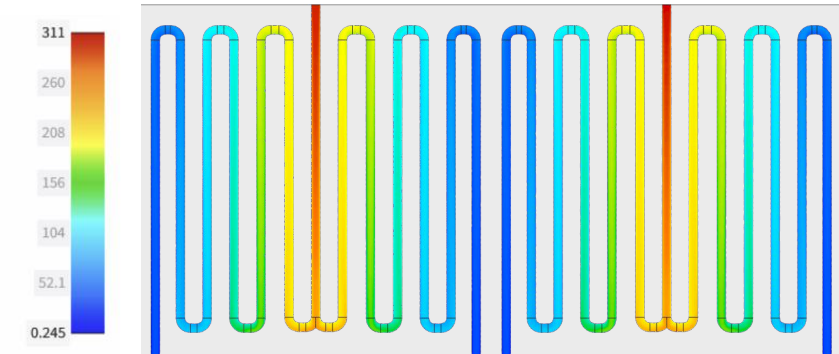
Temperature °C



The improved temperature plot shows a **more even thermal distribution** across the plate.

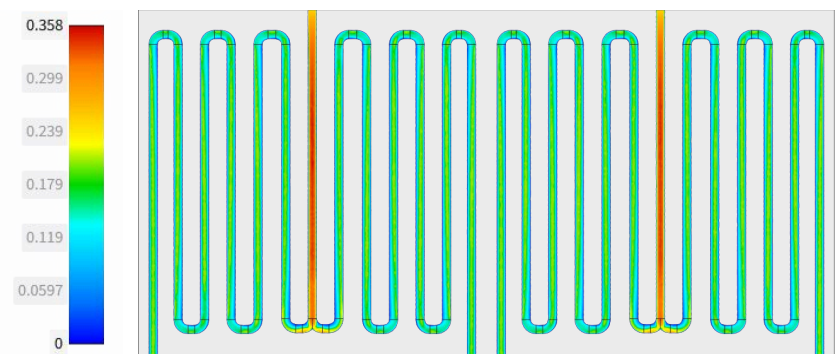
Compared with the earlier design, the **cooler regions** are less concentrated in one area and are instead **spread more broadly** across the channel network.

Total Pressure Pa



The pressure drop increased from **80.8 Pa** to **297 Pa** due to the added bends and smaller channel diameter. At **0.012 kg/s**, this only increases hydraulic power by about **0.0026 W**, so the penalty is very small. This trade-off was accepted to improve temperature uniformity and reduce battery temperature.

Velocity m/s

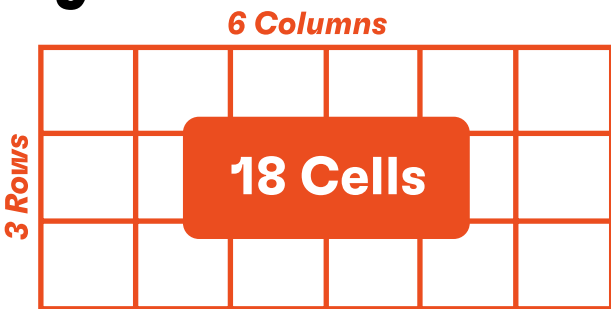


The velocity plot shows a **similar flow pattern** to the original, but the **two-inlet** design creates two local high-velocity regions instead of one.

The maximum velocity increases from **0.219 m/s** to **0.358 m/s**, likely due to the smaller channel diameter.

Cell Temperature Analysis

The CFD treated the battery as a single solid, so it can't show how temperature varies between individual cells, which is important for performance and ageing. To get around this, the volume was split into 18 regions to represent cells across the battery pack.



Temperature data was extracted using a **Python** script run through the ANSYS scripting function. The battery volume was sampled using a **40 × 20 × 10** grid, creating a CSV file with 8000 temperature values across the battery pack.

A second **Python** script grouped these points into the **18 cells** and calculated the mean temperature of each.

The mean temperature is used to give each region a single value, so the continuous CFD results can be treated like separate cell groups. In real batteries, performance and ageing depend on the overall temperature of a cell, so the average is a better way to describe how each region behaves.

Cell Mean Temperature

(X:0-70mm)	Cell 1 40.96°C	Cell 2 39.52°C	Cell 3 41.09°C	Cell 4 41.11°C	Cell 5 39.58°C	Cell 6 41.08°C
(X:70-140mm)	Cell 7 40.83°C	Cell 8 39.34°C	Cell 9 40.96°C	Cell 10 40.96°C	Cell 11 39.40°C	Cell 12 40.96°C
(X:140-210mm)	Cell 13 41.11°C	Cell 14 39.78°C	Cell 15 41.22°C	Cell 16 41.24°C	Cell 17 39.83°C	Cell 18 41.21°C
	(Y:0-70mm)	(Y:70-140mm)	(Y:140-210mm)	(Y:210-280mm)	(Y:280-350mm)	(Y:350-420mm)

ΔT : 1.9 °C

Min: 39.34 °C

Max: 41.24 °C

The cell temperature spread is **1.9 °C**, well within the typical **<5 °C** limit for avoiding uneven ageing. This shows the design keeps temperatures uniform across the pack, reducing the risk of cells ageing at different rates and helping to maintain lifespan and performance.

Conclusion

The end is near

Methods and Modelling

Hand calcs were useful for identifying the key factors that affect the cooling.

Such as: Channel Diameter Flow Velocity Pressure Drop

These guided some of the decisions and gave a good understanding of the system and its behaviour.

However, the accuracy of the hand calcs depend heavily on assumed values and simplifications, meaning the results are only indicative. Small changes in these assumptions can lead to large errors so they are not totally reliable.

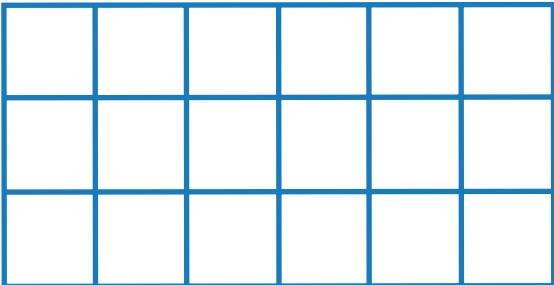
The CFD was much more effective for developing and optimising the design. Was much faster iteration and made it very easy to assess the temperature and the pressure drop together, rather than separately.

With current modern computing, it is much quicker to improve a design through the simulations rather than by adjusting the hand calculations. The CFD is far more practical.

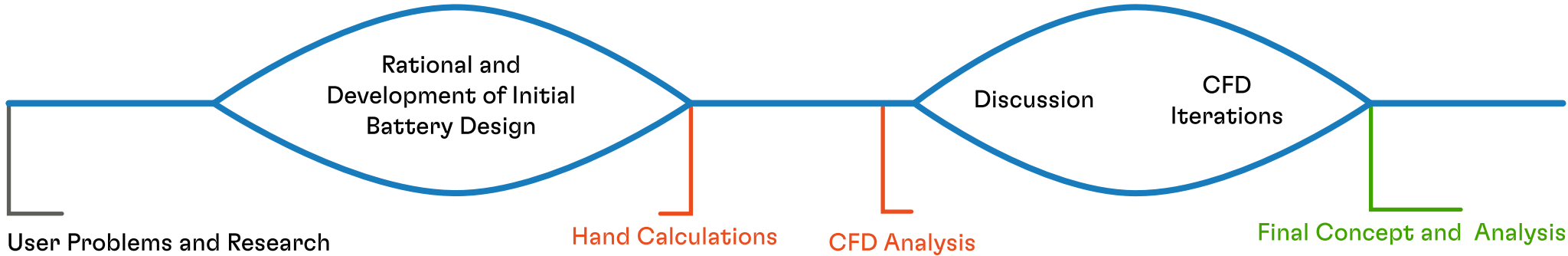
The tools are not easy to use without prior experience (or tutorials).



Using a cell based analysis for the final design did make it more realistic. It better represents how real heat behaviour works rather than treating the battery as a single homogeneous solid. The model assumed 18 identical prismatic cells, which is still a simplification, a real battery packs have much more complex internal structures, like vertical heat pipes or cylindrical cells.



Double Diamond for Battery Design Process

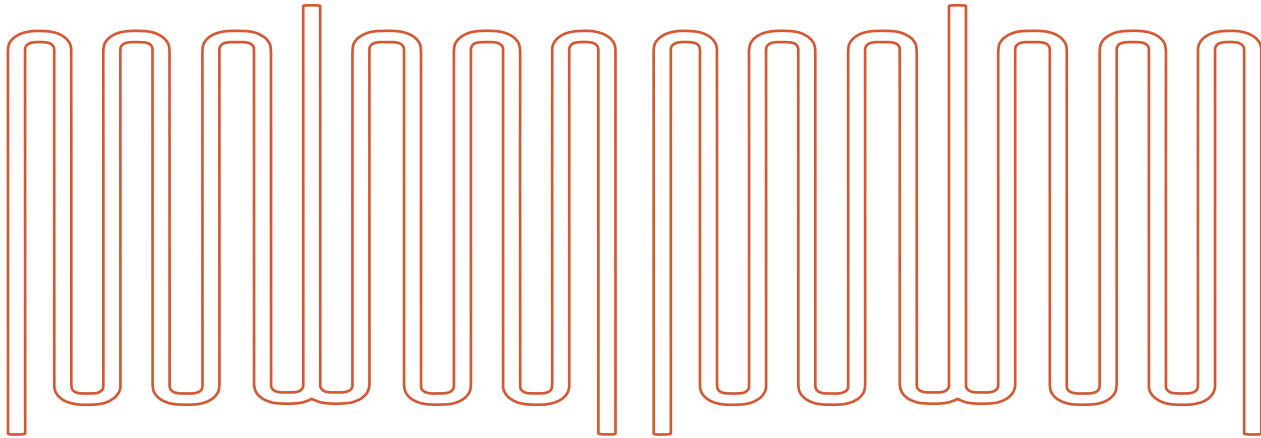


Final & Improvements

The final design has good thermal performance and temperature uniformity and it also maintained a fairly low pressure drop.

There are no significant pressure drops and the temperature difference between the cells remained low showing how the cooling was effective across the pack.

ΔT: 1.9 °C
Drop: 297 Pa



The improvements that were made during the iteration process generally reduced the temperature but increased the pressure drop. This shows the trade-off between improving battery cooling and increasing pressure drop.

Initial Design
Average Temp: 42.4°C
Max Temp: 47.3°C
Pressure Drop: 80.8Pa

Final Design
Average Temp: 42 °C ▼
Max Temp: 45.2°C ▼
Pressure Drop: 297Pa ▲

The design is still not fully optimised and so further improvements could focus on different channel geometries.

The design could also look into using flatter channel shapes instead of circular ones to increase contact area. This would potentially improve heat transfer while also maintaining the flow velocity.

The plate thickness could be reduced which might improve thermal performance. However this could create issues structurally and cause potential durability issues.

References

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1. Murdoch R. Thermofluids Individual Report. Imperial College London; 2026 Feb p. 1–12.
2. Jones M. Thermofluids Individual Report. Imperial College London; 2026 Feb p. 16–20.
3. Hazeri K, Childs P. Thermofluids Spring 2026 Module Handbook. Imperial College London; p. 11–17.
4. Hazeri K. Drag & CFD Fundamentals Thermo-fluids (Spring 25). Imperial College London; 2025.
5. T1 and T2 Wind Tunnels | Faculty of Engineering | Imperial College London [Internet]. www.imperial.ac.uk. Available from: <https://www.imperial.ac.uk/aeronautics/research/facilities/t1-andt2/>
6. Imperialwindtunnel.com. 2026 [cited 2026 Mar 18]. Available from: <https://imperialwindtunnel.com/>
7. Dymott G. How does a Wind Tunnel work? [Internet]. AirShaper. 2023. Available from: <https://airshaper.com/blog/how-does-a-wind-tunnel-work>
8. What is vapor smoothing? | Protolabs Network [Internet]. Protolabs Network. 2024. Available from: <https://www.hubs.com/knowledge-base/vapor-smoothing/>
9. Ultimate Materials Guide - Tips for 3D Printing with PVA [Internet]. Available from: <https://www.simplify3d.com/resources/materials-guide/pva/>
10. Rehm B, Schubert J. Reynolds' Number - an overview | ScienceDirect Topics [Internet]. Sciencedirect.com. 2008. Available from: <https://www.sciencedirect.com/topics/engineering/reynolds-number>
11. Comparison of lithium battery performance at 45°C and other temperatures [Internet]. large-battery.com. 2025 [cited 2026 Mar 18]. Available from: <https://www.large-battery.com/blog/45c-and-lithium-battery-performance-comparison/>
12. Akhtari Zavareh P, Matam AN, Shah K. Heterogeneous Aging in a Multi-cell Lithium-ion Battery System Driven by Manufacturing-induced Variability in Electrode Microstructure: A Physics-Based Simulation Study. Energy Advances. 2025;
13. Li X, Wang S. Flow field and pressure loss analysis of junction and its structure optimization of aircraft hydraulic pipe system. Chinese Journal of Aeronautics. 2013 Aug;26(4):1080–92.
14. Wu G, Liu F, Li S, Luo N, Liu Z, Li Y. Research on Performance Optimization of Liquid Cooling and Composite Phase Change Material Coupling Cooling Thermal Management System for Vehicle Power Battery. Journal of Renewable Materials. 2023;11(2):707–30.
15. Khan N. Simulating and Optimizing an Electric Vehicle Battery Cold Plate [Internet]. Simscale.com. SimScale; 2022 [cited 2026 Mar 18]. Available from: <https://www.simscale.com/blog/simulating-optimizing-electric-vehicle-battery-cold-plate/>
16. Shetty DD, Venugopal V, R P, Zuber M, Anjum Badruddin I, Kini C. Computational flow analysis of different streamline cooling plates for thermal management of lithium-ion battery. Shukla SK, editor. Cogent Engineering. 2022 Mar 14;9(1).
17. Hazeri K. 1D Hand Calculation for Battery Cooling Tutorial. Imperial College London;
18. Hooper WB. The two-K method predict. Chemical Engineering; 1981 Aug.
19. Fusion 360 | 3D CAD, CAM, CAE & PCB Cloud-Based Software | Autodesk [Internet]. www.autodesk.com. Available from: <https://www.autodesk.com/products/fusion-360>
20. Ansys Discovery | 3D Product Simulation Software [Internet]. www.ansys.com. Available from: <https://www.ansys.com/products/3d-design/ansys-discovery>
21. Shah RK, London AL. Laminar Flow Forced Convection in Ducts. Academic Press, New York.; 1978.
22. Hazeri K. Thermo-Fluids: Energy and Design. Imperial College London; 2022 Sep.
23. PYTHON. Python [Internet]. Python.org. Python.org; 2025 [cited 2026 Mar 18]. Available from: <https://www.python.org/>
24. Advantages of Prismatic Cells in EVs: What Are They and How Do They Work - Ennovi [Internet]. Ennovi. 2024 [cited 2026 Mar 18]. Available from: <https://ennovi.com/advantages-of-prismatic-cells-in-evs-what-are-they-and-how-do-they-work/>